

T5) $f \sim R(\theta, 2\theta)$

$$p(x) = \frac{1}{\theta} \mathbb{I}_{[\theta, 2\theta]}$$

$$\mu_f = \int_{\theta}^{2\theta} \frac{1}{\theta} x dx = \frac{1}{\theta} \left(\frac{4\theta^2}{2} - \frac{\theta^2}{2} \right) = \frac{3\theta}{2}$$

$$\mu_f^2 = \int_{\theta}^{2\theta} \frac{1}{\theta} x^2 dx = \frac{1}{\theta} \left(\frac{8\theta^3}{3} - \frac{\theta^3}{3} \right) = \frac{7\theta^2}{3}$$

$$D_f = \mu_f^2 - (\mu_f)^2 = \frac{7\theta^2}{3} - \frac{9\theta^2}{4} = \frac{\theta^2}{12}$$

a) М-г моментов.

$$\alpha_1(\theta) = \bar{x}$$

$$\frac{3\theta}{2} = \bar{x} \Rightarrow \underline{\bar{\theta}} = \frac{2\bar{x}}{3}$$

М-г правдоподобия.

$$(\theta \leq x \leq 2\theta)$$

$$L(\theta) = \left(\frac{1}{\theta}\right)^n \mathbb{I}_{\{\theta \leq x_i \leq 2\theta\}}$$

$$\begin{cases} x_i \geq \theta \\ x_i \leq 2\theta \end{cases}$$

$$\Rightarrow \theta \in \left[\frac{1}{2} x_{\max}, x_{\min} \right]$$

$$\sup L(\theta) = \frac{1}{2} x_{\max} = \underline{\bar{\theta}_2}$$

б) Метод Ньютона: $\forall \theta \in \Theta \quad \mu_{\bar{\theta}} = \theta$

$$\bar{\theta}_1 = \frac{2\bar{x}}{3} = \frac{2}{3} \cdot \frac{1}{n} \sum_{i=1}^n x_i$$

$$\mu(\bar{\theta}_1) = \mu\left(\frac{2\bar{x}}{3}\right) = \frac{2}{3n} \sum \mu x_i = \frac{2}{3n} \cdot n \cdot \mu_f = \frac{2}{3} \cdot \frac{3\theta}{2} = \theta$$

Ньютона

$$\bar{\theta}_2 = \frac{1}{2} x_{\max}$$

$$\frac{1}{2} x_{\max} \sim \frac{1}{2} \frac{(F(x))^n}{\psi(x)}$$

$$F(x) = \int_{\theta}^x p(t) dt = \int_{\theta}^x \frac{1}{\theta} dt = \frac{x-\theta}{\theta}$$

$$\psi(x) = \psi'(x) = \frac{1}{2} \cdot n \cdot (F(x))^{n-1} \cdot F'(x) = \frac{n}{2} \left(\frac{x-\theta}{\theta} \right)^{n-1} \cdot \frac{1}{\theta}$$

$$\mu\left(\frac{1}{2} x_{\max}\right) = \int_{\theta}^{2\theta} \frac{n}{2} \left(\frac{x-\theta}{\theta} \right)^{n-1} \cdot \frac{1}{\theta} x dx =$$

$$\frac{n}{2\theta} \cdot \frac{1}{\theta^{n-1}} \int_0^{2\theta} \frac{(x-\theta)^{n-1}}{t} \cdot x \, dt \quad \left(\begin{array}{l} t = x - \theta \\ x = t + \theta \end{array} \right)$$

$$\begin{aligned} & \frac{n}{2\theta^n} \left(\int_0^\theta t^{n-1} t \, dt + \int_\theta^{2\theta} t^{n-1} \theta \, dt \right) = \frac{n}{2\theta^n} \left(\frac{\theta^{n+1}}{n+1} + \theta \frac{\theta^n}{n} \right) = \\ & = \frac{n}{2\theta^n} \left(\frac{\theta^{n+1} n + \theta^{n+1} (n+1)}{(n+1)n} \right) = \frac{n}{2\theta^n} \theta^{n+1} \left(\frac{n+n+1}{n(n+1)} \right) = \frac{1}{2} \theta \left(\frac{2n+1}{n+1} \right) = \\ & = \theta \left(\frac{2n+1}{2n+2} \right) - \text{смешу} \end{aligned}$$

$$\tilde{\theta}_2' = \frac{1}{L} x_{\max} \cdot \frac{2n+2}{2n+1} - \text{смешу}$$

$$\mathcal{D}\tilde{\theta}_1 = \mathcal{D}\left(\frac{2x}{3}\right) = \frac{4}{9} \mathcal{D}\left(\frac{1}{n} \sum x\right) = \frac{4}{9n^2} \cdot n \cdot \mathcal{D}\xi = \frac{4}{9n} \cdot \frac{\theta^2}{12} \xrightarrow{n \rightarrow \infty} 0$$

- сходимость по гл. условию

$$\mathcal{D}\tilde{\theta}_2' = \mathcal{D}\left(\frac{2n+2}{2n+1} \cdot \frac{1}{L} x_{\max}\right) = \left(\frac{2n+2}{2n+1}\right)^2 \cdot \left(\frac{1}{L}\right)^2 \cdot \mathcal{D}(x_{\max})$$

$$\begin{aligned} M x_{\max} &= \int_\theta^{2\theta} x^2 \cdot n \cdot \left(\frac{x-\theta}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} \, dx = n \cdot \frac{1}{\theta} \cdot \frac{1}{\theta^{n-1}} \int_\theta^{2\theta} x^2 \cdot (x-\theta)^{n-1} \, dx = \\ &= \frac{n}{\theta^n} \int_0^\theta (t+\theta)^2 \cdot t^{n-1} \, dt = \frac{n}{\theta^n} \left(\int_0^\theta t^2 \cdot t^{n-1} \, dt + \right. \\ &+ \int_0^\theta 2\theta \cdot t \cdot t^{n-1} \, dt + \int_0^\theta \theta^2 \cdot t^{n-1} \, dt \Big) = \frac{n}{\theta^n} \left(\frac{\theta^{n+2}}{n+2} + \right. \\ &+ 2\theta \cdot \frac{\theta^{n+1}}{n+1} + \theta^2 \cdot \frac{\theta^n}{n} \Big) = \frac{n\theta^2}{n+2} + \frac{2n\theta^2}{n+1} + \theta^2 \end{aligned}$$

$$\begin{aligned} \mathcal{D}(x_{\max}) &= \frac{n\theta^2}{n+2} + \frac{2n\theta^2}{n+1} + \theta^2 - \left(\theta \left(\frac{2n+1}{n+1} \right) \right)^2 = \dots \\ &= \frac{\theta^2 n}{(n+1)^2 (n+2)} \end{aligned}$$

~~но гл. условие~~

$$\mathcal{D}\tilde{\theta}_2' = \frac{1}{4} \cdot \frac{\theta^2 n \cdot (2n+2)^2}{(2n+1)^2 (n+1)^2 (n+2)} \xrightarrow{n \rightarrow \infty} 0 - \text{сходимость по гл. условию}$$

б) $\frac{1}{27n}$

$$> \frac{(2n+2)^2 \cdot n}{4(2n+1)^2 (n+1)^2 (n+2)} = \frac{n}{(2n+1)(n+2)}, \quad n > 2$$

~~$\tilde{\theta}_2'$~~ $\tilde{\theta}_2'$ $\tilde{\theta}_1$

2) Готный довер интервал

$$f(\bar{x}_n, h) = \frac{x_{\max}}{\theta} \quad (\text{нормировка})$$

$$P(f(\bar{x}_n, h) < t) = P\left(\frac{x_{\max}}{\theta} < t\right)$$

$$P(x_{\max} < t\theta) = (F(t\theta))^n = \left(F\left(\frac{t\theta + \theta}{\theta}\right)\right)^n = \left(\frac{\theta(t+1)}{\theta}\right)^n = (t+1)^n$$

$$t_1 = q_{1-\beta} \quad , \quad t_2 = q_{\frac{\beta}{2}}$$

$$F(t_1) = \frac{1-\beta}{2}$$

$$\Rightarrow (t_1 + 1)^n = \frac{1-\beta}{2}$$

$$\Rightarrow t_1 = 1 + \sqrt[n]{\frac{1-\beta}{2}}$$

$$F(t_2) = \frac{\beta}{2}$$

$$\Rightarrow (t_2 + 1)^n = \frac{\beta}{2}$$

$$\Rightarrow t_2 = 1 + \sqrt[n]{\frac{\beta}{2}}$$

$$P\left(1 + \sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{\max}}{\theta} < 1 + \sqrt[n]{\frac{\beta}{2}}\right) = \beta$$

$$\boxed{\frac{x_{\max}}{1 + \sqrt[n]{\frac{1-\beta}{2}}} < \theta < \frac{x_{\max}}{1 + \sqrt[n]{\frac{\beta}{2}}}}$$

9) Асимптот. довер интервал

$$\bar{\theta}_1 = \frac{2\bar{x}}{3} = \frac{2}{3} \bar{x}_1$$

$$\nabla g(x) = \frac{2}{3}$$

$$K_{11} = x_{11} - x_1 x_1$$

$$\sqrt{n} \frac{g(\bar{x}) - g(x)}{\sigma(\bar{x})} \sim N(0,1), \quad \bar{\theta} = g(\bar{x})$$

$$\sigma^2(x) = \nabla^T g(x) K \nabla g(x) = \frac{2}{3} (x_2 - x_1^2) \cdot \frac{2}{3} = \frac{4}{9} (x_2 - x_1^2)$$

$$\frac{\sqrt{n}(\bar{\theta}_1 - \theta)}{\frac{2}{3} \sqrt{x_2 - x_1^2}} \sim N(0,1)$$

$$t_1 = u_{1-\beta} \quad , \quad t_2 = u_{\frac{\beta}{2}}$$

$$P\left(t_1 < \frac{\sqrt{n}(\bar{\theta}_1 - \theta)}{\frac{2}{3} \sqrt{x_2 - x_1^2}} < t_2\right) = \beta$$

$$\Rightarrow \boxed{\bar{\theta}_1 - \frac{2}{3} \frac{\sqrt{x_2 - x_1^2}}{\sqrt{n}} t_2 < \theta < \bar{\theta}_1 - \frac{2}{3} \frac{\sqrt{x_2 - x_1^2}}{\sqrt{n}} t_1}$$